From figure matches to paper links

Modality-Specific and Relation-Aware Modeling for Scientific Figure Linking

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0.6096

V8 Kaggle hidden-test macro-F1

0.3641 in the initial submission

01 The linking task

Given two figures or captions from astronomy papers, predict one relation from the pair's content.

Same figure
Exact match, for image–caption pairs only

Same paper
Both objects originate from one paper

Related papers
One source paper directly cites the other

Unrelated papers
No preceding relation holds

10,000 train / 10,000 hidden test
Per split: 3,000 CC · 3,000 II · 4,000 IC

02 Three routes through the final system

Frozen pretrained models supply features or scores. Downstream heads learn from labeled pairs.

PAIR TYPE	EVIDENCE	DECISION
CC **Caption + caption**	**SigLIP2 text, SPECTER2, NeuML** astronomy embeddings + lexical overlap	→ **XGBoost** Three paper relations
II **Image + image**	**SigLIP2 So400m (384 × 384)** XGBoost + learned neural projection	→ **Blend citation score only** Three paper relations
IC **Image + caption**	**Qwen3-VL 4B/8B gate** Blend first-token YES/NO scores in log-odds space	→ **Pass threshold** **Same figure** **Otherwise** **Three paper relations** So400m + SPECTER2, XGBoost, citation blend

Symmetric embedding features: absolute difference, elementwise product and cosine similarity.

03 Kaggle progression

Hidden-test macro-F1 over four classes

V1	Frozen SigLIP2	0.3641
V2	Scientific-text features	0.4472
V3	**Qwen3-VL IC routing**	0.5660
V4	So400m visual features	0.5843
V5	IC blend and gate calibration	0.5919
V6	4B/8B gate ensemble	0.5982
V7	Astronomy features for CC	0.6081
V8	**Neural specialist for II**	0.6096

+0.1188 at V3

The largest step between submissions came when the exact-match gate was introduced.

V1 to V8 total gain +0.2455

04 Local ablations

Caption + caption / pooled five-fold OOF

0.6071 → 0.6388

Adding NeuML to the existing caption features improved macro-F1 in all five folds.

Exact-match gate / separate validation

Binary F1 on 400 held-out image–caption pairs

Qwen3-VL 4B	0.8911
Qwen3-VL 8B	0.8785
4B + 8B ensemble	0.9163

Image + image / projection trade-off

On pooled OOF predictions, the standalone neural head raised direct-citation F1 (0.3454 → 0.4405) but lowered macro-F1 (0.4994 → 0.4917). The score blend added 0.0015 on Kaggle.

05 Direct citation errors

Predictions for 1,000 true citation-linked pairs in each local OOF diagnostic.

Same paper · Related · Unrelated

II · So400m XGBoost

275	310	415

IC · Residual XGBoost only

317	323	360

Each classifier missed most citation-linked pairs. Both confused them with same-paper and unrelated pairs.

THE LIMIT OF PAIR CONTENT

A caption can describe a figure precisely. A citation may depend on context outside both objects.

How to interpret the results

Pair-level validation
Out-of-fold (OOF) splits were stratified by pair. Figure, caption and paper overlap was unmeasured. Performance on unseen papers remains unresolved.

Model selection
Choosing the CC variant and tuning the II blend on OOF predictions can make local gains look larger than they would on new data.

Leaderboard feedback
Kaggle kept test labels private, so we could not test whether score differences between submissions were statistically significant.

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In the published paper: Figure 1 and Tables 1–4, 6, 8 and 10